

LieOptics: A Ray-Free Framework for Optical Modeling and Analysis

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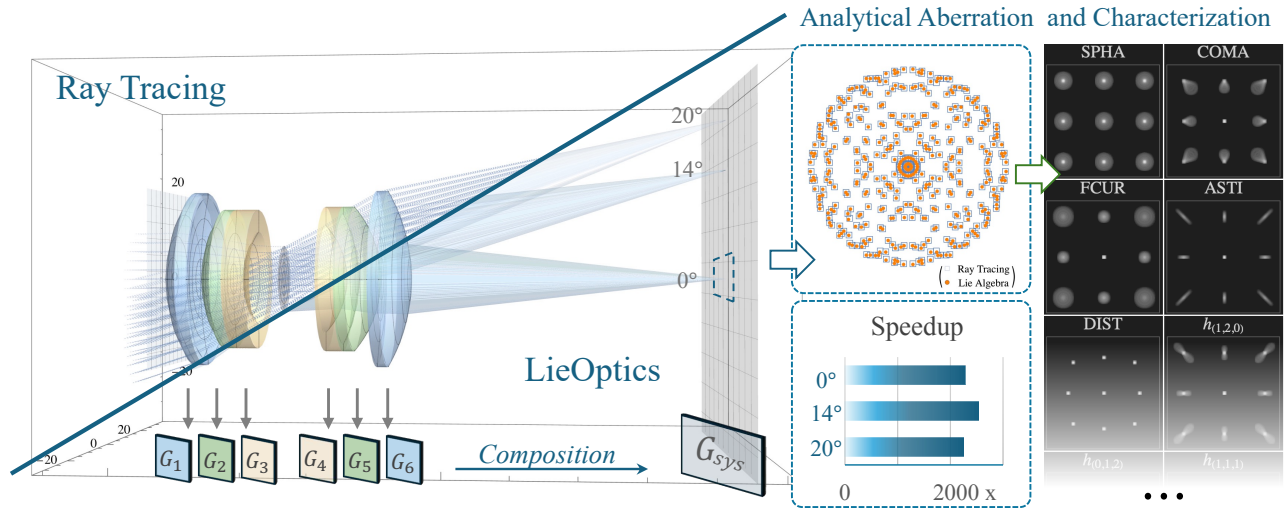


Fig. 1. **LieOptics Framework** demonstrated on a Double Gauss lens [Ohashi 1999]. **(Left)** While ray tracing (RT) propagates individual rays independently, LieOptics models the evolution of the entire ray bundle analytically. Each optical element is encoded as a Lie operator G_i , and the full system is obtained by composing them into a single G_{sys} . **(Center)** The composed operator directly yields the aberration spot pattern across field angles (0° , 14° , 20°), overlaid against RT for validation, with over $1000\times$ speedup compared to RT. **(Right)** LieOptics decomposes the system aberration by type: spherical (SPHA), coma (COMA), field curvature (FCUR), astigmatism (ASTI), distortion (DIST), and up to 9th order analytically, a breakdown inaccessible to RT, which is limited to 3rd-order Seidel aberrations via post-hoc fitting.

Most optical design pipelines, including today’s commercial tools, rely on ray tracing, where system performance is evaluated through aggregate metrics such as spot diagrams or wavefront error. While effective for predicting overall image quality, these metrics offer limited insight into how individual elements and optical surfaces contribute to aberrations, making diagnosis and correction largely heuristic. At a fundamental level, aberrations arise from the collective deformation of ray bundles, not individual rays. However, ray tracing samples this behavior point-wise and obscures the underlying ray bundle structure, thereby limiting interpretability. We introduce LieOptics, which models optical elements as Lie-algebraic analytic operators acting directly on ray bundles. These operators explicitly encode aberration type and order, and compose naturally across elements to yield a single system-level operator. This approach provides direct access to the full aberration structure, enabling per-element attribution, sensitivity analysis, and principled aberration diagnosis. LieOptics generalizes ray transfer matrices from paraxial approximations to higher order aberrations for the first time. By operating on analytic representations instead of heuristically sampled rays, we demonstrate a speedup of over $1000\times$ compared to conventional ray tracing modeling and analysis.

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CCS Concepts: • **Computing methodologies** → **Computational photography**.

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1 Introduction

Optical modeling underlies modern imaging systems, from smartphone cameras to AR/VR headsets and scientific instruments. Yet after decades of progress, the core modeling, analysis, and design workflows remains fundamentally built upon ray tracing. In this paradigm, optical systems are analyzed by simulating individual rays, and in particular, how they refract through refractive surfaces. Characterizing a lens system therefore requires intensive ray casting, often involving millions of rays per simulation [Sun et al. 2021]. As a result, optical prototyping and optimization are formulated as iterative parameter refinements through repeated forward simulations. This workflow underpins every commercial platform, such as Zemax and CODEV, and modern differentiable and end-to-end pipelines [Tseng et al. 2021; Wang et al. 2021; Yang et al. 2024a].

Despite its success, ray tracing operates as a black-box forward simulation. It produces a post-hoc evaluation, such as spot diagrams or wavefront errors, that reflects the behavior of sampled rays at the image plane. All the aberrations from elements are folded into one image pattern with no direct mapping from lens parameters to aberration structure. In fact, ray tracing provides only limited insights into per-element aberration attribution, exhibiting only the lowest-order decomposition by aberration type. Without access to this fine-grained aberration structure, optical modeling remains a simulation-guided process that depends on carefully chosen initial designs, iterative trial-and-error refinement, and the expertise of experienced engineers.

Recent differentiable design pipelines reduce manual intervention by replacing iterative simulation with gradient-based optimization [Côté et al. 2023; Wang et al. 2022; Yang et al. 2024a], but inherit the same limitation: the forward model maps lens parameters to aggregate quality metrics; the aberration structure behind those metrics stays hidden, and the contribution of each element remains obscure. As a result, structurally distinct parameter configurations of lens systems can produce nearly identical aggregate performance [Antonov et al. 2023; Côté et al. 2023], and prototyping remains guided by heuristics rather than structured understanding.

These limitations share a common origin: ray tracing models light transport one ray at a time; a single ray obeys Snell’s law only in isolation. But aberration is fundamentally the *collective behavior of a ray bundle*. It emerges when rays from an object point fail to converge to a common image point. The natural object to model is therefore the ray bundle, described as a distribution over ray positions and directions in phase space [Torre 2005]. In this representation, optical elements act as operators that transform the entire ray bundle in phase space. For instance, a first order phase space transformation represents paraxial optics that transforms the rays linearly, and higher order transformations encode aberrations beyond the paraxial approximation. However, existing treatments either remain limited to the paraxial regime or are restricted to Seidel aberrations of a single refracting surface [Barion et al. 2022; Navarro-Saad and Wolf 1986], and they do not provide access to higher-order aberration structures for a lens system.

In this paper, we introduce LieOptics, an analytical framework for optical modeling and prototyping. We build analytical representations of lens systems and ray bundles, where each optical element is modeled as a Lie-algebraic operator acting on the ray bundle in phase space. Composing these operators across a multi-element system yields a single compound operator whose polynomial terms expose the system’s full aberration analytically as explicit functions of lens parameters. As a result, the evaluation cost of our framework scales with system complexity rather than ray count. Importantly, aberrations in LieOptics arise directly from the operator structure rather than as post-hoc artifacts inferred from ray statistics.

This operator representation unlocks capabilities previously inaccessible to ray tracing. Direct access to per-element operator structure enables aberration attribution, i.e., identifying which elements contribute which aberration types and at what order. The analytical form supports sensitivity analysis, revealing how individual lens parameters affect each aberration term. Aberration budgeting becomes possible just by inspection, allowing designers to allocate and

trade off contributions across elements before optimization. LieOptics achieves over 1000× speedup compared to ray tracing, enabling real-time prototyping and large-scale lens system modeling.

The contributions of this work are:

- *First analytical operator-based pipeline to map lens parameters directly to aberrations.* LieOptics analytically maps lens parameters to closed-form aberration coefficients for compound lenses, without sampling individual rays.
- *Aberration attribution per element up to 9th order.* Lie operator naturally models aberrations by element and by type, enabling direct diagnosis of which elements contribute to which aberration.
- *Aberration decomposition and budgeting.* The analytical structure reveals how individual parameters affect each aberration and allows for allocating aberration contributions across elements before optimization.
- *Over 1000× speedup over ray tracing.* LieOptics achieves orders-of-magnitude improvements in computational and memory efficiency, enabling real-time evaluation and large-scale comparison of candidate designs.

2 Related Works

Classical Optical Modeling and Lens Design. Classical optical modeling is built on ray tracing, where individual rays are propagated through refractive surfaces according to Snell’s law, and system performance is evaluated from their distribution at the screen [O’Shea and Bentley 2018; Zemax 2011]. In this paradigm, which forms the foundation of all major platforms including Zemax and CODEV, design optimization proceeds by iteratively perturbing lens parameters against a scalar merit function, typically RMS spot size [O’Shea and Bentley 2018; Yang et al. 2024a]. Since the forward model produces no structured aberration representation, optimization is prone to converge to local minima, and its performance is heavily dependent on the choice of initial design and engineering expertise [Gao et al. 2025]. Global search methods have been proposed to broaden exploration, including saddle point methods with automatic element insertion [Zoric et al. 2024], genetic algorithms [Höschel and Lakshminarayanan 2018], large language model-guided search [Zoric et al. 2025], and data-driven approaches that search patent databases or generate initial designs from learned distributions [Côté et al. 2021; Sun et al. 2015]. These methods expand design space coverage but remain bottlenecked by ray tracing speed and memory, and provide no structured access to aberration.

Computational Optics and Imaging. Computational methods have accelerated optical simulation and rendering and scaled them to complex systems, but remain limited by ray tracing. Differentiable ray tracing enables gradient-based optimization of optical systems by replacing manual iteration with automatic differentiation [Sun et al. 2021; Teh et al. 2022; Tseng et al. 2021; Wang et al. 2021]. Recent work extends these methods to joint optimization with neural image reconstruction networks, enabling task-specific designs for extended depth-of-field, large field-of-view, and high dynamic range imaging [Teh et al. 2024; Yang et al. 2024b]. Curriculum-based approaches [Yang et al. 2024a] and discrete-continuous optimization [Teh et al. 2025] further broaden the design space accessible

to gradient methods. Population-based pipelines such as OptiFusion [Gao et al. 2025] combine simulated annealing, genetic algorithms, and gradient descent to explore diverse initial configurations. However, these end-to-end frameworks inherit the same representational limitation as ray tracing. The forward model maps lens parameters to scalar quality metrics, whereas the aberration structure behind those metrics remain hidden, and no error can be traced to an individual element. As a result, structurally distinct configurations can produce nearly identical performance [Antonov et al. 2023; Côté et al. 2023], and optimization remains heuristic.

Polynomial optics replaces costly render-time ray tracing through lens systems with truncated ray-transfer polynomials [Hanika and Dachsbacher 2014; Hullin et al. 2012; Schrader et al. 2016]. While this makes rendering efficient, obtaining the system polynomial requires composing per-element polynomials, which can cause intermediate expressions to grow rapidly, or fitting the polynomial to ray-traced data. Moreover, these representations also do not guarantee the ability to attribute aberrations to individual optical elements and conservation of ‘etendue’ at every order of ray transfer [Sadeghi and Sarisaman 2026; Torre 2005]. In contrast, LieOptics guarantees both properties by its very construction.

Aberration Theory. Classical aberration theory, including Seidel’s third-order theory [Seidel 1856] and Hopkins’ wave aberration formulation [Hopkins 1950], established the foundation for analyzing optical aberrations. However, these methods are largely limited to low-order aberrations. Although Buchdahl extended the aberration theory to higher orders [Buchdahl 1954], his approach still requires ray tracing and does not provide a direct mapping from lens parameters to aberrations, limiting its use for optical modeling. Lie algebraic methods were introduced for accelerator and beam physics [Dragt 1982; Navarro-Saad and Wolf 1986; Torre 2005], and were shown to represent optical transfer maps as Lie algebraic operators [Sanchez Mondragon and Wolf 1986]. While prior work based on Lie operators is limited to paraxial systems or single surfaces, recent work has extended the approach to third-order aberrations in simple compound systems [Barion et al. 2022]. Our work develops the first Lie-operator framework for general compound-lens modeling, generalizing the ray transfer matrix beyond paraxial optics. We derive per-element operators and closed-form rules for composing them into a complete lens system. The resulting model captures aberrations up to 9th order and is over 1000× faster than ray tracing.

3 The LieOptics Framework

We introduce the phase-space description of ray bundles, whose evolution is governed by the optical Hamiltonian and solved by a Lie operator (Section 3.1). Next, we represent each optical element as a Lie operator determined by the lens parameters, such as surface curvature and refractive index (Section 3.2). These operators provide the aberrations contributed by each element in closed form, without the dense sampling that ray tracing (RT) requires. Then, we compose the per-element operators into a single system-level operator to model the action of the entire system on rays (Section 3.3). This composition produces the optical system aberrations while keeping

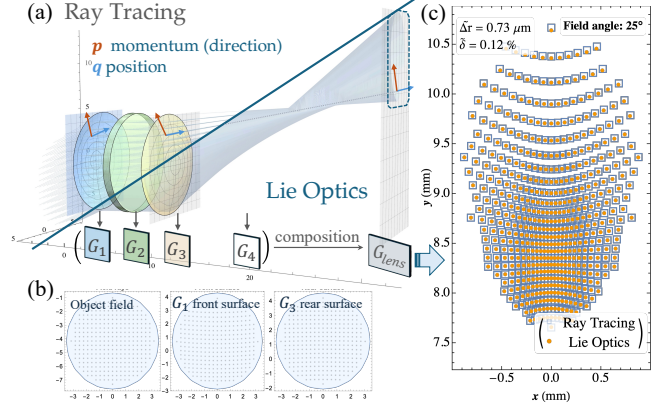


Fig. 2. Comparison between ray tracing (RT) and LieOptics. (a) We model a ray bundle through a thick lens, where a ray is parameterized by its transverse position q and momentum (direction) p in phase space. LieOptics decomposes the lens into optical elements (refractive surfaces G_1 , G_3 and free medium G_2 , G_4), each represented as a Lie operator encoding the aberrations it introduces. The full lens is then expressed as a single composed operator G_{lens} . (b) RT (gray dots) must sample rays intensively at each surfaces to capture aberrations. In contrast, LieOptics captures the evolution of the entire ray bundle (blue shaded cross-section) analytically through operators, while offering aberration contributions per element and per type up to 9th order. (c) G_{lens} maps the bundle from the blue reference plane to the image plane in closed form, giving the aberrated spot pattern directly (dashed region). We sample this pattern to validate against RT baseline: the median spot error is $\Delta r = 0.73 \mu\text{m}$ and the RMS relative error is $\delta = 0.12\%$.

element-level aberration contributions identifiable. Finally, we derive performance metrics such as the RMS spot size, together with the aberration type and contribution of each element directly from the composed system operator (Section 3.4). Detailed derivations are available in the Supplementary Material.

3.1 Ray Bundles in Phase Space and the Lie Operator

Aberration is the departure of a ray bundle from its ideal focus, as illustrated in Figure 2. Because this deviation accumulates as the ray bundle propagates through successive optical surfaces, the ray bundle provides a natural object for aberration modeling. Here, we introduce a ray-bundle representation that analytically characterizes the aberrations of the complete lens system while simultaneously attributing them to individual optical elements.

We parameterize each ray in a ray bundle jointly by its optical momentum $\mathbf{p} = (p_x, p_y)$ and transverse position $\mathbf{q} = (q_x, q_y)$, as indicated by the red and blue arrows in Figure 2. The optical momentum here is the unit ray direction scaled by the local refractive index. This parameterization is formally identical to the 4D light field (2D position $\otimes$ 2D direction) in computer graphics [Levoy and Hanrahan 1996]. This formulation describes the unique trajectory of each ray without dense ray sampling. The four-dimensional space $\mathbf{u} = (\mathbf{p}, \mathbf{q})$, which constitutes the optical phase-space [Dragt and Finn 1976; Torre 2005], enables us to analytically track how the ray bundle is deformed at each element.

A ray bundle that propagates through an ideal thin lens imaging system gradually converges to a sharp focus. Such an ideal system can be described by the ABCD matrix, which models the paraxial propagation as a linear map on rays. However, the ray bundle fails to converge to a point in a practical thick lens and shows aberrations that manifest as a spot pattern as shown in Figure 2. To capture this nonlinear map of aberrations, we apply Hamilton's equations:

$$\frac{\partial \mathbf{u}}{\partial z} = -\{H(\mathbf{p}, \mathbf{q}; z), \mathbf{u}\}, \quad (1)$$

where the Hamiltonian $H(\mathbf{p}, \mathbf{q}; z) = \sqrt{n(\mathbf{q}, z)^2 - \mathbf{p}^2}$ encodes the local refractive index n of each element, and z denotes the propagation distance. The Poisson bracket $\{H, \mathbf{u}\}$ is the generator that captures the ray bundle evolution by changing the phase-space coordinates:

$$\hat{H} \mathbf{u} = \{H, \mathbf{u}\} = \partial \mathbf{u} / \partial \mathbf{p} \partial H / \partial \mathbf{q} - \partial \mathbf{u} / \partial \mathbf{q} \partial H / \partial \mathbf{p}.$$

Exponentiating this generator gives the Lie operator $G(H) \equiv e^{-z\hat{H}}$, a nonlinear map that transforms the entire ray bundle in a single step, directly encoding aberration in its structure rather than inferring it from ray sampling. Based on this, LieOptics (LO) generalizes the ABCD matrix formalism for off-axis aberrations and systematically extends it to multi-element systems for the first time.

We obtain the operator G in closed form by truncating the nonlinear map at a finite polynomial degree. In our work, we derive the operator up to 10th degree; the same construction extends to higher orders. Notably, this polynomial truncation is symplectic at every order [Dragt 1982]. Therefore, *étendue of the ray bundle is preserved by construction* and the truncation adds no artifacts.

3.2 Optical Elements as Parameterized Operators

We now identify the Lie operator for each element by determining its Hamiltonian in phase space $(\mathbf{p}, \mathbf{q})$. As aberrations arise from nonlinear ray transfer, we expand the Hamiltonian around the optical axis as a sum of homogeneous polynomials, and organizes the Hamiltonian terms:

$$H(\mathbf{p}, \mathbf{q}) = \sum_{k=1} H_k = H_1 + H_2 + H_3 + \dots, \quad (2)$$

where each H_k is a $2k$ -degree polynomial, and generates aberrations of order $2k - 1$. The Hamiltonian H_1 of the paraxial regime generates the linear small-angle ray transfer, close to the optical axis, describing the paraxial ABCD matrix; H_2 generates the 3rd-order (Seidel) aberrations, and so on. Note that H_k terms directly encode the aberrations from the lens parameters of an optical element.

The actions of H_k generators do not commute, i.e., their associated Lie operators cannot be applied in an arbitrary order. The Dragt-Finn factorization [Dragt 1982] provides a structured way to represent the full operator as an ordered sequence of simple factors:

$$G(H) = G(H_5) \cdot G(H_4) \cdot G(H_3) \cdots G(H_1). \quad (3)$$

The factor $G(H_1)$ describes the paraxial behavior of the optical system, i.e., ideal linear propagation and imaging close to the reference ray. The remaining factors $G(H_k)$ ($k \geq 2$) produce successive nonlinear corrections, or aberrations. Each factor accounts for actions at a particular order, with the higher-order factors describing progressively finer deviations from ideal paraxial behavior.

The order of the factorization is essential. Since the underlying generators do not commute: applying $G(H_4)$ before $G(H_3)$, for example, does not produce the same result as applying in the opposite order. The Dragt-Finn form preserves these order-dependent interactions while separating the basic paraxial transport from the system's aberrations. This separation is essential for aberration modeling and truncation of Lie operators in closed form.

With the operator structure established, we next determine the vector of aberration coefficients $\mathbf{h}_k$ describing each H_k polynomial directly from the lens parameters. For an axially-symmetric lens as illustrated in Figure 2, rotational symmetry restricts H_k to a finite basis of rotationally-invariant monomials in $(\mathbf{p}, \mathbf{q})$, i.e., $(\mathbf{p} \cdot \mathbf{p})(\mathbf{p} \cdot \mathbf{q})^2, \dots, (\mathbf{p} \cdot \mathbf{p})^{k_1}(\mathbf{p} \cdot \mathbf{q})^{k_2}(\mathbf{q} \cdot \mathbf{q})^{k_3}$; the generator H_k is identified by its coefficient vector $\mathbf{h}_k$ in this basis, where $k = k_1 + k_2 + k_3$.

Each lens consists of two components: (1) a homogeneous medium of index n and thickness d that propagates the bundle in position while leaving the direction unchanged; (2) refractive surfaces that redirect rays in non-uniform directions while keeping the position unchanged and are the main source of aberrations. We derive the coefficients $\mathbf{h}_k$ for the lens by composing surface transformations [Navarro-Saad and Wolf 1986] and matching the overall result term-by-term against the Lie operator expansion. Complete derivations are in the Supplementary Material.

The aberration coefficients of each component thus follow directly from their parameters in closed form. For instance, the third-order coefficients of a refractive lens can be expressed analytically as:

$$\mathbf{h}_2 = \left\{ -\frac{d}{8n^3}, -\frac{d(1-n)}{2n^3R_1}, \frac{(n-1)}{4n^3R_1R_2} (d(1-n)R_2 + n^2R_1(R_1 - R_2)), \right. \\ \left. -\frac{d(n-1)^2}{2n^3R_1^2}, -\frac{d(1-n)^3}{2n^3R_1^3} + \frac{1-n}{2nR_1^2} + \frac{n-1}{2R_2^2} \right. \\ \left. + \frac{(n-1)(n(2dR_1 - 1) + 1)}{2nR_1R_2}, \dots \right\} \quad (4)$$

where R_1, R_2, n describe the front and rear surface radii and the glass' refractive index. Here, the first five coefficients describe the five aberration types: spherical aberration (SPHA), coma, field curvature (FCUR), astigmatism (ASTI), and distortion (DIST) for a thick lens (the last term in this example is omitted since it does not affect ray position). Higher-order coefficients $\mathbf{h}_k$ ($k = 3, 4, 5$) are derived similarly. Each component, therefore, corresponds to an aberration type, also obtained directly from the lens parameters in closed form.

To model aspherical surfaces, we utilize the sag equation:

$$S(q) = \frac{c_1 q^2}{1 + \sqrt{1 - c_1^2 q^2}} + c_2 q^4 + c_3 q^6 + c_4 q^8 + c_5 q^{10} + \dots, \quad (5)$$

where q is the ray height, $c_1 = 1/R$ is the vertex curvature of the base sphere with radius R , and $c_2, c_3, \dots$ are the aspherical coefficients that add higher-order corrections. Expanding Equation (5) in q shows that each c_l contributes to aberration coefficients $\mathbf{h}_k$ ($k \geq l$).

3.3 System Composition and Aberration Attribution

To describe a multi-element system, we compose per-element operators into a system one $G_{\text{lens}} = G_1 \cdot G_2 \cdots G_N$. This concatenation

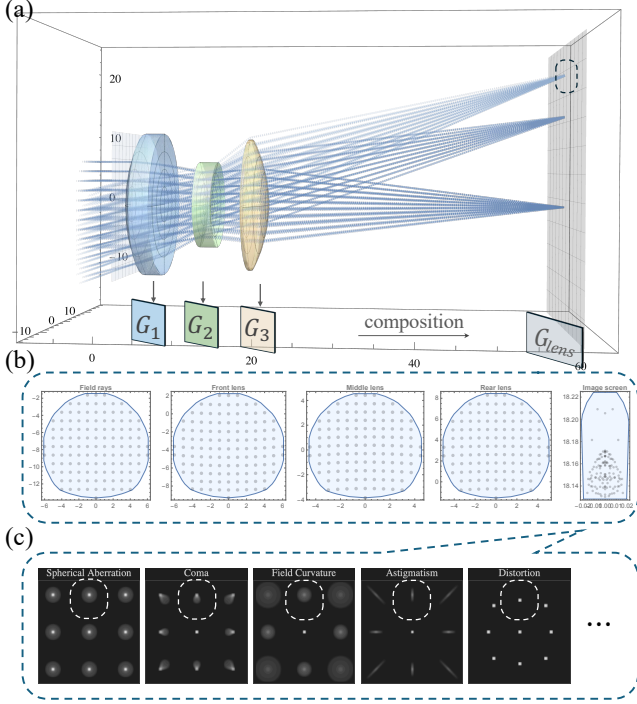


Fig. 3. LieOptics modeling of a Triplet lens [Geiser 2016], with per-element operators G_1, G_2, G_3 composed into G_{lens} . (a) The ray bundle is propagated through each element analytically. (b) Ray bundle cross-sections (blue shades) tracked per element from input to image screen; the dashed box shows the final spot pattern for 20° field rays from infinity, agreeing with RT. (c) LieOptics decomposes the on-screen pattern by aberration type (spherical, coma, field curvature, astigmatism, distortion, ...) up to 9th order, which is limited to Seidel (3rd-order) aberrations by RT.

produces the system-wide aberration and the per-element aberration attribution. The key challenge is that each element's aberration is defined in its own local reference frame, so the operators cannot be composed directly. Even completely paraxial propagation between elements shifts and scales the ray bundle. As a result, an aberration introduced at one element is expressed in different phase-space coordinates when it arrives at the next element. So we transport aberrations of each element to a common reference frame. In our implementation, we use the plane adjacent to the first lens surface (blue plane in Figure 2) as the reference plane.

Algebraically, we transport an aberration generator H by the adjoint action of the operator $\exp(\Lambda)$:

$$\hat{H}' = \text{Ad}_{\exp(\Lambda)} \hat{H} = \exp(\Lambda) \cdot \hat{H} \cdot \exp(-\Lambda), \quad (6)$$

where Λ is the paraxial generator of the preceding optical elements. Every Hamiltonian generator H_k of element i is transported to the front reference plane using the accumulated paraxial operators of all preceding elements, $\exp(\Lambda) = \exp(\Lambda_1) \cdots \exp(\Lambda_{i-1})$. For order k , $\exp(\Lambda)$ is a matrix built from the ABCD matrices of the previous elements, transforming the lens coefficient vector $\mathbf{h}_k$ to the vector $\mathbf{h}'_k$ at the front reference plane. We derive these matrices for arbitrary aberration order in the Supplementary Material. Once we obtain all

aberration coefficients on the reference plane, we combine them via the Baker-Campbell-Hausdorff (BCH) formula [Hall 2015]:

$$e^{\hat{H}_{\text{composed}}} = e^{\hat{H}_{\text{front}}} \cdot e^{\hat{H}_{\text{rear}}}. \quad (7)$$

The aberration coefficients $\mathbf{h}_{\text{front/rear}}$ of the two elements' generators determine the composed coefficients $\mathbf{h}_{\text{composed}}$ directly.

The composition rule works directly on the coefficients $\mathbf{h}$, with no intermediate expressions and no traced rays. We apply it element by element to construct the system operator G_{lens} . Third-order coefficients add directly across elements, while fifth-order and beyond pick up cross-terms from Poisson-bracket couplings $\{H_{\text{front}}, H_{\text{rear}}\}$ between lower-order aberrations of different elements; see Supplementary Material. So each element's contribution remains as a labelled, separable term in the analytical system-wide coefficient vector $\mathbf{h}_{\text{sys},k}$. The contribution of each element can be identified directly, for example, how much spherical aberration is introduced by element (3), or how much of the coma from element (4) is canceled by element (5). Ray tracing must recover the same information indirectly by perturbing each element separately, requiring one experiment per element and aberration type. In contrast, LieOptics composition provides this information directly as a byproduct of the same algebra used to generate the spot diagram.

This per-type aberration attribution provides another important insight into optical modeling. Aberrations introduced by different elements often partially cancel, and the LieOptics composition reveals whether such cancellation is structural or accidental. A cancellation is *structural* when it follows analytically from the lens parameters and holds across a broad class of aberration types, so it persists as parameters change. In contrast, an *accidental* cancellation occurs when the numerical contributions happen to offset for one aberration type at one configuration, so it generally does not persist under perturbation. Thus, the analytical composition provides a fine-grained description of where individual aberrations originate and how contributions from different elements interact.

For instance, the Lie operator G_{lens} of the triplet in Figure 3 produces the same spot diagram as ray tracing while additionally exposing the per-type contribution of each element. In this way, LieOptics composition reveals not only which aberrations cancel but also the mechanism behind their cancellation; such information is not directly accessible from ray tracing alone.

3.4 Direct Aberration Metrics

We derive the aberration metrics directly from the composed system-level operator. Since aberration is the deviation from the ideal image point, subtracting that point from the action of G_{lens} on a ray bundle $(\mathbf{p}, \mathbf{q})$ gives the transverse displacement $\Delta Q(\mathbf{p}, \mathbf{q})$ at the image plane. Therefore, the RMS spot radius for each field point can be obtained as the root-mean-squared displacement of the transverse coordinate $\Delta Q(\mathbf{p}, \mathbf{q})$ integrated over the reference plane Ω :

$$\sigma_{\text{rms}} = \sqrt{\int_{\Omega} |\Delta Q(\mathbf{p}, \mathbf{q})|^2 \mathbf{p} d\mathbf{p}}. \quad (8)$$

Note that $(\mathbf{p}, \mathbf{q})$ here denotes the ray direction and ray coordinates on the front reference plane. Because the displacement is polynomial,

the above integral admits a closed-form solution. We can therefore evaluate the metric directly, without sampling rays.

4 Results

We evaluate LieOptics on multiple lens systems, including the double Gauss (Figure 1) and triplet (Figure 3) as accuracy benchmarks, and the thick lens (Figure 2), fisheye-spherical, and fisheye-aspherical as stress tests at large field of view (FoV). All LieOptics predictions are computed in closed form, while ray tracing (RT) provides only the baseline spot diagrams. We further demonstrate per-element and per-aberration-type attribution, sensitivity analysis, and an ablation of truncation order. We refer the reader to the Supplementary Material for details and additional evaluations.

Implementation. Algorithm 1 models lens systems in two stages: the first stage composes the system’s Lie operator G_{lens} from the lens parameters (steps 1–7), and the second one produces the truncated ray-transfer polynomials for a ray bundle (step 8). We implement the LieOptics framework in Python 3.12 to time the full-pass evaluation, and additionally in Mathematica 14.1 for the RT baseline, as it provides the efficient numerical solver FindRoot to trace rays. All tests are run on a single CPU core of a MacBook Pro (M4, 16 GB RAM).

Algorithm 1 LieOptics optical modeling pipeline

Require: $\mathcal{L} = [\ell_1, \dots, \ell_N]$, where ℓ_i stores the parameters of lens i

Ensure: aspherical parameters are stored distinctively

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1: for  $i \leftarrow 1$  to  $N$  do
2:    $\Lambda_i, \hat{H}_i \leftarrow$  get paraxial and aberration generators of lens  $i$ 
3:    $\hat{H}'_i \leftarrow \text{Ad}_{\exp(\Lambda_i)} \hat{H}_i$   $\triangleright$  transport to reference plane, Eqn. 6
4: end for
5:  $\Lambda_{\text{sys}} \leftarrow \Lambda_N \cdots \Lambda_2 \Lambda_1$   $\triangleright$  compose paraxial ABCD matrices
6:  $\mathbf{h}_{\text{sys},k} \leftarrow \text{BCH}(\mathbf{h}'_{1,k}, \dots, \mathbf{h}'_{N,k})$   $\triangleright$  compose aberration coefficients
7:  $G_{\text{sys}} \leftarrow G(H_5) \cdots G(H_2) \cdot G(\Lambda)$   $\triangleright$  Assemble operator, Eqn. 3
8: return  $G_{\text{sys}}(\mathbf{p}, \mathbf{q})$   $\triangleright$  Lens operator applied to ray coordinates

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4.1 Accuracy and Feasibility of Aberration Modeling

The composed operator G_{lens} analytically predicts the aberration coefficients and the deformation of the ray bundle. For benchmarking, we derive the 9th-order ray-transfer polynomials and compare their spot diagrams against ray tracing. We evaluate accuracy in two regimes. When the spot is small and comparable to the diffraction limit, we compute RMS_{err} as the RMS distance between the rays corresponding to LieOptics and RT. When the spot grows significantly larger, the diffraction limit is no longer a meaningful reference. We instead report the median ray error Δr and the RMS per-ray relative error δ , which measure each ray’s LO–RT deviation by its distance from the RT spot centroid $\bar{q}$:

$$\delta = \langle (\|\mathbf{q}_i^{\text{LieOptics}} - \mathbf{q}_i^{\text{RT}}\| / \|\mathbf{q}_i^{\text{RT}} - \bar{\mathbf{q}}\|)^2 \rangle^{1/2} \times 100\%. \quad (9)$$

Figure 4 shows the accuracy test on the 40° FoV triplet lens [Geiser 2016] at 0°, 14°, and 20°, with $\text{RMS}_{\text{err}} = 0.027, 0.802, \text{ and } 4.817 \mu\text{m}$. The first two errors stay well below the diffraction limit and the largest error at 20° remains comparable to this limit, confirming

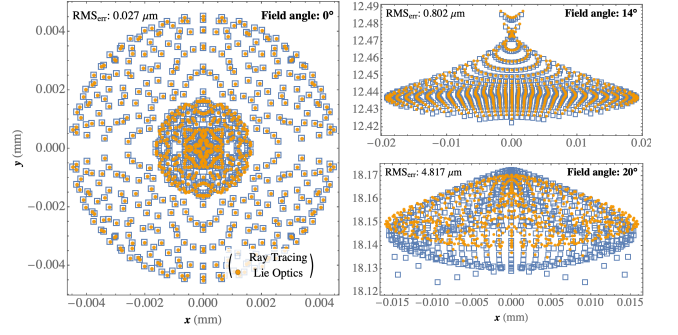


Fig. 4. Accuracy of LieOptics against RT on the Triplet lens [Geiser 2016] (see Figure 3). Spot diagrams shown at three field angles 0°, 14°, 20°, LieOptics predictions (orange) overlap with RT (blue squares). The per-angle RMS_{err} measures the RMS deviation between the two spot patterns. At all three field angles, the error remains well below or comparable to the diffraction limit of $2.9 \mu\text{m}$, confirming its accuracy across wide off-axial conditions.

the accuracy of LieOptics across the FoV of the triplet. Figure 5 demonstrates the accuracy evaluation on the 40° FoV Double Gauss lens [Ohashi 1999], giving $\text{RMS}_{\text{err}} = 0.016, 0.065, \text{ and } 0.978 \mu\text{m}$ at the same angles, all well below the diffraction limit. This 6-element system test shows the accuracy of LieOptics across the full FoV of this lens.

4.2 Aberration Attribution

LieOptics attributes aberrations to individual optical elements in closed form. Each generator of a Lie operator is associated with an aberration coefficient, naturally exposing the contribution of each aberration type to each element. For example, a 9th-order operator contains up to 52 aberration coefficients per element, all computed without perturbing the optical system. Figure 7 shows this attribution for the triplet, where each bar represents an element’s signed contribution to one aberration type. Lens 2 dominates across all aberration types at both 3rd and 5th order. We therefore analyze

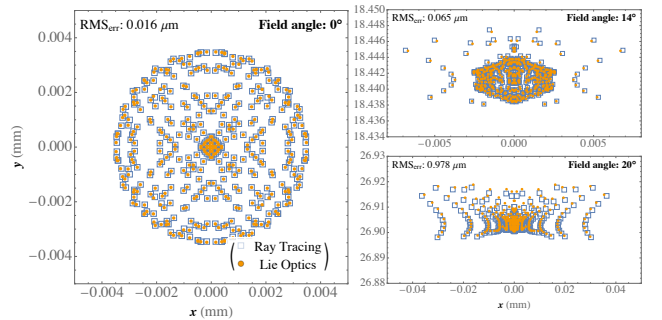


Fig. 5. Accuracy of LieOptics on Double Gauss lens [Ohashi 1999] (Figure 1). On-screen spot diagrams at three field angles 0°, 14°, 20°, overlaying LieOptics predictions (orange) against RT (blue squares). The RMS_{err} at all three field angles remains well below the diffraction limit of $2.9 \mu\text{m}$, validating the analytical accuracy of the composed operator G_{sys} .

Lens 2 further to identify which of its parameters can reduce the system aberration.

Figure 8 shows the sensitivity analysis for the triplet. A parameter sweep shows sensitivity to two Lens 2 parameters: the front surface radius $R_{2,\text{front}}$ and the thickness t_2 . The two parameters show opposite trends. Moving $R_{2,\text{front}}$ away from its nominal value reduces individual aberrations (COMA, FCUR, and ASTI) but increases the RMS spot size, trading off the total image quality. In contrast, increasing t_2 reduces the RMS spot size monotonically while increasing each aberration type. These trends reveal the trade-off between correcting individual aberrations and minimizing the overall spot size. Because LieOptics evaluates each sweep in closed form without finite differences, the analysis remains interactive in real-time prototyping.

We analyze the Double Gauss in Figure 9. Lens 3 and Lens 4 sit symmetrically about the aperture stop. Each introduces large aberrations, however, their signed contributions largely cancel at the system level across all aberration types. LieOptics makes such cancellation patterns directly visible and identifies complementary lens pairs without tracing rays. The same signed decomposition supports aberration budgeting: because per-element contributions compose to the system total, we can assign how much aberration each element may introduce or must cancel before optimization begins.

4.3 Speed and Scalability

We evaluate two computational costs of LieOptics framework: composing the system operator (step 1–7 in Algorithm 1), and evaluating rays with the resulting operator (step 8). System composition is performed only once per lens system. For the Double Gauss lens LieOptics composes the complete 9th-order operator with all terms in 0.026 s in Python. The resulting operator can then evaluate ray bundles at any field angle. For per-ray evaluation, LieOptics is over 1000× faster than RT (Figure 1) at each field angle.

This speedup comes from the closed-form Lie algebraic representation. Once the system operator is composed, LieOptics evaluates each ray directly using the truncated ray-transfer map. Its cost therefore scales only with the number of rays as $\mathcal{O}(N_{\text{rays}})$. In contrast, RT samples rays and computes ray-surface intersections at every lens element, resulting in a cost that scales as $\mathcal{O}(N_{\text{rays}}N_{\text{lens}})$. Therefore, LieOptics scales efficiently to multi-element lenses.

4.4 Stress Test of Wide Field Angle Accuracy

To test accuracy at large field angles, we evaluate LieOptics on three wide-FoV systems: a thick lens (singlet in Figure 2), a 11-elements fisheye spherical lens, and a fisheye aspherical lens where the aspherical design further reduces the spot size [Schrade et al. 2016]. The spot sizes of these systems are larger than their diffraction limits, so we adopt the median ray error Δr and the RMS per-ray relative error δ for the stress tests.

First, we test the highly aberrated thick lens (front/rear radius $|R| = 15$ mm, refractive index $n = 1.4$, and thickness $d = 2.5$ mm). Reaching at a field angle of 25° (half of the 50° FoV), we measure a median LO–RT ray error of $\Delta r = 0.73 \mu\text{m}$, below the diffraction limit. Since the spot spans more than 3 mm vertically, we measure

Table 1. Accuracy of LieOptics against RT on the fisheye-spherical lens with a diffraction limit ($3.0 \mu\text{m}$). Agreement holds up to 30° field angle (half of 60° FoV), then degrades as the error of 9th-order truncation grows.

Field angle	0°	7°	14°	20°	25°	30°	35°	40°
δ (%)	0.78	0.68	0.70	0.51	1.76	12.49	70.00	338.56
Δr (μm)	0.91	0.39	0.31	1.38	6.00	22.20	107.91	1206.42

$\delta = 0.12\%$. This indicates that LieOptics remains accurate across a 50° FoV even on a singlet with severe aberrations.

Next, we evaluate LieOptics on two 11-element fisheye lenses. Figure 6 and Table 1 measure the spherical variant, where LieOptics agrees with RT up to a 60° FoV, with a median ray error comparable to the diffraction limit.

The fisheye-aspherical lens adds aspherical corrections to four of its surfaces, which control its spherical aberration. However, under the same 9th-order truncation, we measure the error $\Delta r = 21.13, 15.64, 30.03$, and $65.29 \mu\text{m}$ at field angles of $0^\circ, 7^\circ, 14^\circ$, and 20° , an order of magnitude above the spherical variant. The loss of accuracy stems from how LieOptics models aspherical corrections: the aspherical curvatures c_l enter the system operator at the matching polynomial degree l . A 10th-order coefficient reaches only the 9th aberration order, with higher orders uncounted in this work. This is not an intrinsic limitation: [Schrade et al. 2016] show that a 15th-degree ray map models this lens accurately across the full 180° FoV, so a higher truncation order should recover the same accuracy.

4.5 Ablation Test

The accuracy of LieOptics relies on higher aberration orders. We compose the operator once, truncate the ray-transfer polynomial further from 9th order (70 terms) to the paraxial ABCD map (2 terms), and re-evaluate the same bundle on three lenses (Supplementary material).

The error falls by roughly a factor of five each time we truncate the map by two higher orders. On the thick lens at 25° , $\Delta r = 105.21, 20.36, 3.82$, and $0.73 \mu\text{m}$ at orders 3, 5, 7, and 9, and the fisheye-spherical lens at 20° falls from 101.70 to $1.38 \mu\text{m}$; the Double Gauss lands every ray within the diffraction limit from 7th order. The spot diagrams show that higher orders extend the agreement from the core of each spot to the whole bundle, which leads to tangible improvement of LieOptics accuracy for wide field angles.

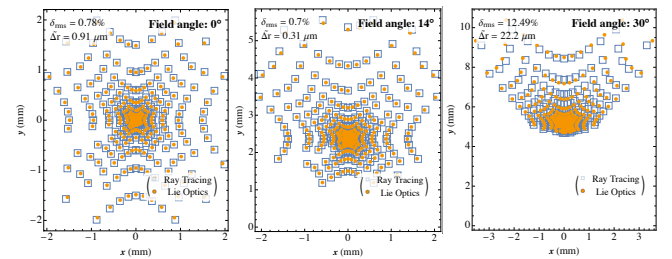


Fig. 6. Spot diagrams of fisheye-spherical at field angles $0^\circ, 14^\circ$, and 30° , overlaying LieOptics predictions (orange) against RT (blue squares); the agreement holds through 14° and degrades at 30° (Table 1).

5 Discussion

LieOptics provides an analytical, ray-free framework for optical modeling, which composes per-element Lie operators into a single system operator. It evaluates spot diagrams and provides per-element and per-type attribution in closed form, at over 1000× the speed of ray tracing. The aberration attribution further uncovers the structured aberrations of a lens system and lets us identify signature lens pairs that expose structural cancellations, which ray tracing can recover partially only through repeated perturbation.

Currently, LieOptics models the geometric ray-transfer map; wave effects such as diffraction, Fresnel transmission, and stray reflections are orthogonal to it and can be handled downstream, as for any ray models. Moreover, LieOptics accommodates spectral dependence by making the refractive index wavelength-dependent, and can extend to anamorphic and freeform surfaces by relaxing rotational symmetry, at the cost of a larger monomial basis. To model aperture stops, we can restrict the phase-space domain of the bundle, which requires modeling the aperture boundary explicitly.

The efficiency of LieOptics opens two directions for future work. We will push the truncation to higher aberration orders, since the deviation of a ray bundle at large off-axial angles is carried by exactly those terms; this should extend sub-diffraction-limit accuracy to wider fields in fisheye lenses. LieOptics also enables differentiable aberration optimization with respect to lens parameters, giving exact gradients for efficient lens design.

Acknowledgments

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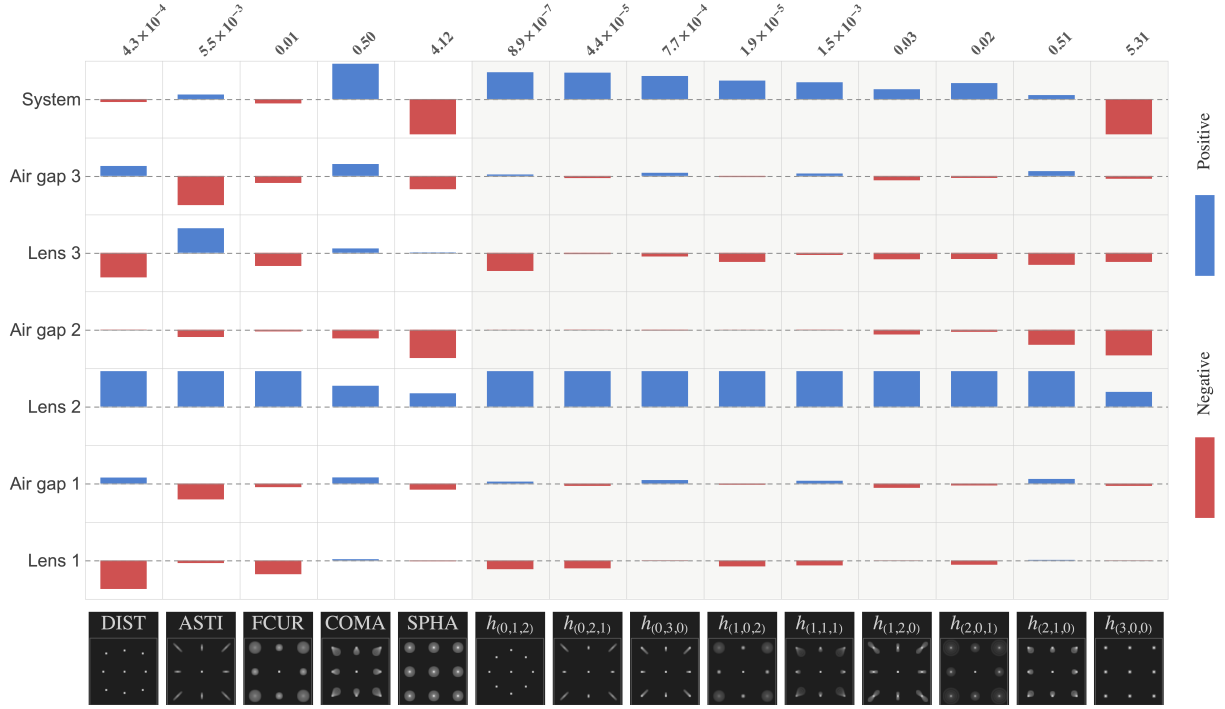


Fig. 7. Per-element and per-type aberration attribution for the Triplet lens [Geiser 2016]. Each row is an optical element (Lens 1–3 and air gaps); each column shows the aberration intensity of a particular type, with its characteristic 9-field-point spot pattern shown at the bottom. Each bar encodes signed aberration intensity (positive value in blue, negative in red), normalized by the aberration value shown at the top. The left unshaded columns show primary Seidel aberrations (3rd order ($n = 2$): DIST, ASTI, FCUR, COMA, SPHA); the right shaded columns cover secondary aberrations (5th order ($n = 3$): h_{m3}). The System row shows the net contribution after all elements cancel or accumulate. We show that lens 2 (middle lens of the triplet) is the single largest contributor across all aberration types and orders, motivating the sensitivity analysis in Figure 8.

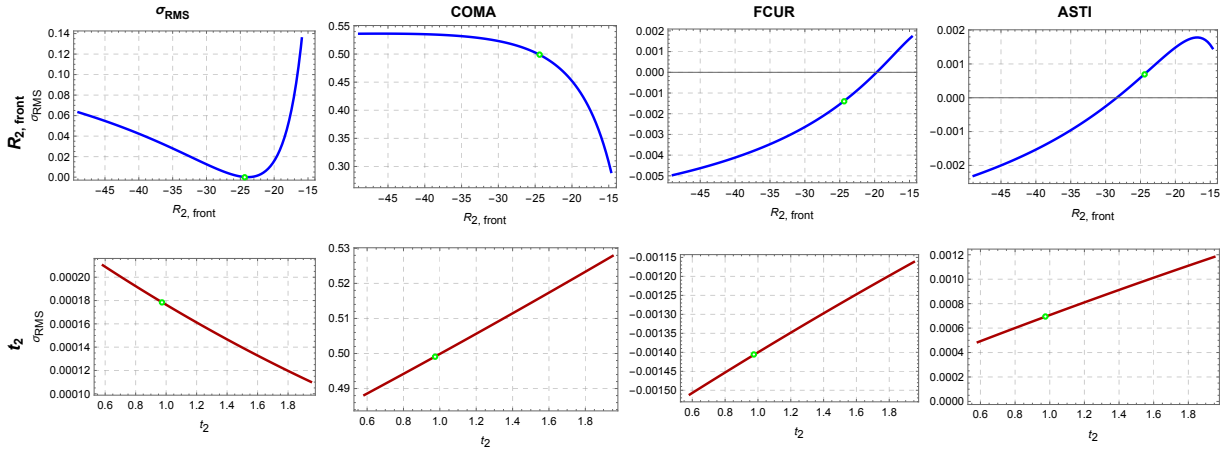


Fig. 8. Sensitivity analysis of the Triplet lens [Geiser 2016]. We identify lens 2 (middle element) as the single largest aberration contributor in Figure 7. Each panel sweeps one parameter against the overall on-axis RMS spot size (σ_{RMS}) and three dominant 3rd-order aberration types (COMA, FCUR, ASTI); the green dot marks the nominal design value. **(Top row)** Sweeping the front surface radius $R_{2,\text{front}}$: moving away from the optimum reduces individual aberration types while the overall spot size increases, revealing a trade-off between per-type correction and total image quality. **(Bottom row)** Sweeping the lens thickness t_2 : the opposite behavior emerges: spot size decreases monotonically while all aberration types increase, showing that thickness trades global blur for higher-order aberration types that blur the image at sensor corners.

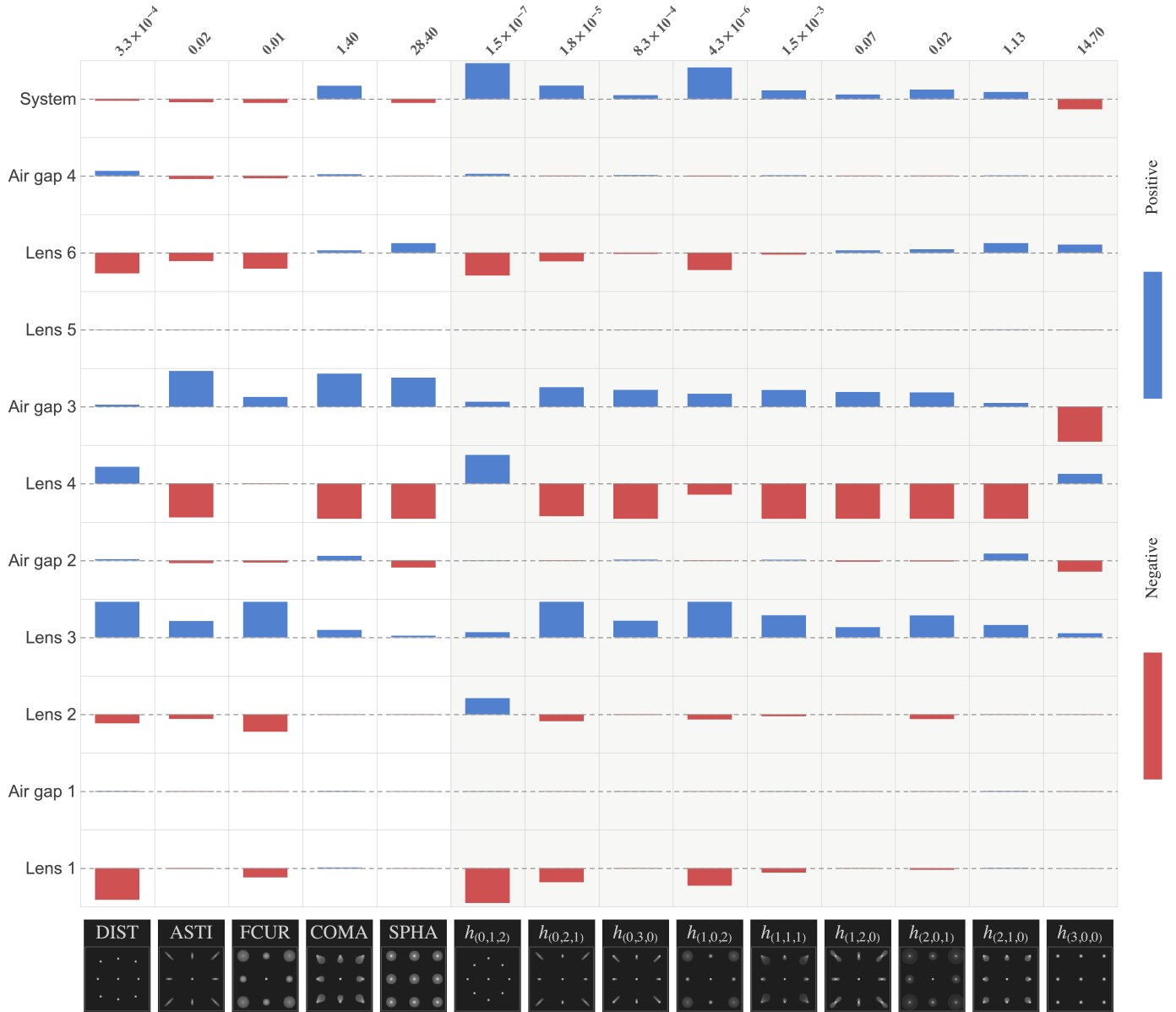


Fig. 9. Aberration attribution by element and type for the Double Gauss lens [Ohashi 1999], using the same visualization as Figure 7. Columns represent aberration types (illustrated by their spot patterns at the bottom) and rows represent optical elements. Blue and red bars indicate positive and negative signed contributions, normalized to the values at the top; the System row gives the net after composition. Primary Seidel aberrations (3rd order, $n = 2$: DIST, ASTI, FCUR, COMA, SPHA) appear in the unshaded columns; secondary aberrations (5th order, $n = 3$: h_{m_3}) in the shaded columns. We show that lens 3 and 4, the two middle elements flanking the aperture stop (Air gap 2), generate sizable aberrations of all types, yet their contributions largely cancel at the system level. This structural cancellation is not coincidental: their symmetric placement about the aperture stop couples their aberration coefficients, producing a robust cancellation that persists under perturbation. LieOptics makes this mechanism directly legible, offering insights for lens prototyping and modeling.